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1. AE 2026

A. Formulae

1.1.1

$$\Pr(M)+\Pr(E)-\Pr(ME)+\Pr(M+E)'=1 \tag{1.1.1.1}$$

1.1.2

$$A+A'=1,\quad AA'=0 \tag{1.1.2.1}$$

$$\implies \Pr(A+A')=\Pr(1)=1\quad \Pr(AA')=\Pr(0)=0 \tag{1.1.2.2}$$

1.1.3

$$\Pr(A+A')=\Pr(A)+\Pr(A') \tag{1.1.3.1}$$

1.1.4 Difference rule of probability

$$\Pr(AB')=\Pr(A)-\Pr(AB) \tag{1.1.4.1}$$

Solution: From (1.1.2.1)

$$A=A\cdot 1=A(B+B')=AB+AB' \tag{1.1.4.2}$$

Also,

$$(AB)(AB')=0 \tag{1.1.4.3}$$

From (1.1.3.1),

$$\Pr(A)=\Pr(AB+AB')=\Pr(AB)+\Pr(AB') \tag{1.1.4.4}$$

Rearranging, we get:

$$P(AB')=P(A)-P(AB) \tag{1.1.4.5}$$

1.1.5 The Laplace transform is defined as

$$g(t)\overset{\mathcal{L}}{\longleftrightarrow}G(s)=\int_0^\infty e^{-st}g(t)\,dt \tag{1.1.5.1}$$

1.1.6

$$g'(t)\overset{\mathcal{L}}{\longleftrightarrow}sG(s)-g(0) \tag{1.1.6.1}$$

Solution: Applying integration by parts,

$$u=e^{-st},dv=g'(t)\,dt \tag{1.1.6.2}$$

$$\implies \int_0^\infty e^{-st}g'(t)\,dt=\left[e^{-st}g(t)\right]_0^\infty-\int_0^\infty\left(-se^{-st}\right)g(t)\,dt \tag{1.1.6.3}$$

$$=-g(0)+s\int_0^\infty e^{-st}g(t)\,dt \tag{1.1.6.4}$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2 G(s) - sg(0) - g'(0) \tag{1.1.7.1}$$

1.1.8

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.8.1}$$

Solution:

$$\implies e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{-st} \left(e^{at} g(t)\right) dt \tag{1.1.8.2}$$

$$= \int_0^\infty e^{-(s-a)t} g(t)\, dt \tag{1.1.8.3}$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \tag{1.1.9.1}$$

1.1.10

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \tag{1.1.10.1}$$

Solution:

$$\int_0^\infty e^{-st}\, dt = \left(-\frac{e^{-st}}{s}\right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \tag{1.1.10.2}$$

1.1.11

$$tu(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s^2} \tag{1.1.11.1}$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

$$\begin{aligned} \cos(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2} \\ \sin(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \end{aligned} \quad \Re(s) > a \tag{1.1.11.2}$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{\jmath at} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-\jmath a}, \quad \Re(s) > a \tag{1.1.11.3}$$

$$\implies \cos(at)u(t) + \jmath \sin(at)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s+\jmath a}{(s-\jmath a)(s+\jmath a)} \tag{1.1.11.4}$$

$$= \frac{s}{s^2+a^2} + \jmath \frac{a}{s^2+a^2} \tag{1.1.11.5}$$

resulting in (1.1.11.2).

1.1.12 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \tag{1.1.12.1}$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \tag{1.1.12.2}$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \tag{1.1.12.3}$$

in (1.1.12.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \tag{1.1.12.4}$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \tag{1.1.12.5}$$

yielding

$$y(t) = \beta \cos(\alpha t) + \frac{\gamma}{\alpha} \sin(\alpha t) \tag{1.1.12.6}$$

from (1.1.11.2).

1.1.13 (Euler's Method of Forward Differences.) Numerically solve

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \tag{1.1.13.1}$$

Solution: By definition,

$$\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{h \to 0} \frac{y(x_0 + h) - y(x_0)}{h} \tag{1.1.13.2}$$

which can be expressed using the notation

$$y'_0 = \frac{y_1 - y_0}{h} \tag{1.1.13.3}$$

Similarly,

$$y''_n = \frac{y'_{n+1} - y'_n}{h} \tag{1.1.13.4}$$

From (1.1.13.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \tag{1.1.13.5}$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.13.6}$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.14.1}$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi n f_0 t}\, dt = \delta_n \tag{1.1.15.1}$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)\,(t - nT_0) \tag{1.1.16.1}$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \tag{1.1.16.2}$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \tag{1.1.16.3}$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.4}$$

Solution: From (1.1.16.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \right) e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.5}$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \tag{1.1.16.6}$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \tag{1.1.16.7}$$

upon substituting from (1.1.16.2) and (1.1.15.1), yielding (1.1.16.4).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}\,(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{1.1.17.1}$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}\,(t), \quad T_0 = 2. \tag{1.1.18.1}$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \tag{1.1.18.2}$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.16.4),

$$c_n = \frac{1}{2}\int_{-1}^{1} e^{-j\pi kt}\,dt \tag{1.1.18.3}$$

$$= \left[\frac{\sin(\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{\frac{1}{2}} \tag{1.1.18.4}$$

The following code verifies the result in Fig. 1.1.18.1. Note that Fig. 1.1.18.1 is continuous while (1.1.18.1) is discontinuous. So (1.1.18.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

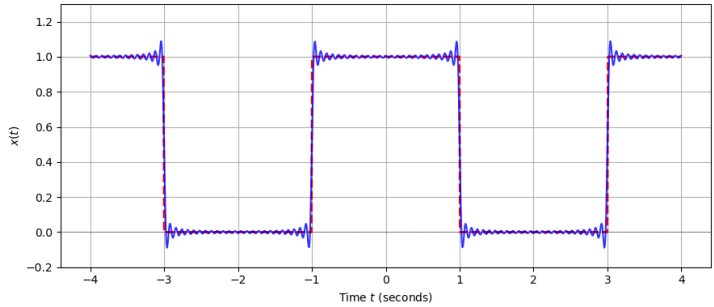


Fig. 1.1.18.1. Gibbs phenomenon

1.1.19 If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} c_n, \tag{1.1.19.1}$$

show that

$$x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} c_n e^{-\mathrm{J}2\pi nt_0}, \tag{1.1.19.2}$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t-t_0)\,e^{-\mathrm{J}2\pi nf_0t}\,dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0(t+t_0)}\,dt \tag{1.1.19.3}$$

$$= e^{\mathrm{J}2\pi nf_0t_0}\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0t}\,dt \tag{1.1.19.4}$$

after a change of variables. Note that the limits of integration are unchanged due the periodicity of $x(t)$. Transferring the exponential term in (1.1.19.4) to the L.H.S yields (1.1.19.2).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.1.20.1}$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.1.20.2}$$

1.1.21 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}\,(A) = 1. \tag{1.1.21.1}$$

where both u and v are n-dimensional real-valued non-zero column vectors.
Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \tag{1.1.21.2}$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.1.21.3}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.21.4}$$

Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.1.21.5}$$

yielding (1.1.21.1).

1.1.22

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right| \tag{1.1.22.1}$$

1.1.23

$$\left|\mathbf{A}\mathbf{B}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.23.1}$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k \tag{1.1.23.2}$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to

verify that

$$\left|\mathbf{EB}\right| = \left|\mathbf{E}\right|\left|\mathbf{B}\right|. \tag{1.1.23.3}$$

$$\therefore \left|\mathbf{AB}\right| = \left|\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.23.4}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.23.5}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right|\left|\mathbf{B}\right| \tag{1.1.23.6}$$

For $\mathbf{B} = \mathbf{I}$,

$$\left|\mathbf{A}\right| = \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right| \tag{1.1.23.7}$$

which upon substituting in (1.1.23.6) yields

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.1.23.8}$$

1.1.24

$$\left|\mathbf{A^{-1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.24.1}$$

1.1.25 If

$$\mathbf{Ax} = \lambda\mathbf{x}, \tag{1.1.25.1}$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.26 Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^{n}\lambda_i = \mathrm{Tr}\left(\mathbf{A}\right) \tag{1.1.26.1}$$

$$\prod_{i=1}^{n}\lambda_i = \left|\mathbf{A}\right| \tag{1.1.26.2}$$

1.1.27 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.
Solution: From (1.1.24.1),

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = \left|\left(\mathbf{A} - \lambda\mathbf{I}\right)^\top\right| \tag{1.1.27.1}$$

$$= \left|\mathbf{A}^\top - \lambda\mathbf{I}\right|, \tag{1.1.27.2}$$

completing the proof.

1.1.28 The equation of a conic with directrix $\mathbf{n}^\top\mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathrm{g}\left(\mathbf{x}\right) = \mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0 \tag{1.1.28.1}$$

where

$$\mathbf{V} = \left\|\mathbf{n}\right\|^2\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.28.2}$$

$$\mathbf{u} = ce^2\mathbf{n} - \left\|\mathbf{n}\right\|^2\mathbf{F}, \tag{1.1.28.3}$$

$$f = \left\|\mathbf{n}\right\|^2\left\|\mathbf{F}\right\|^2 - c^2e^2 \tag{1.1.28.4}$$

Solution:

$$\|\mathbf{x}-\mathbf{F}\|^2 = e^2\frac{(\mathbf{n}^\top\mathbf{x}-c)^2}{\|\mathbf{n}\|^2} \tag{1.1.28.5}$$

$$\implies \|\mathbf{n}\|^2(\mathbf{x}-\mathbf{F})^\top(\mathbf{x}-\mathbf{F}) = e^2\left(\mathbf{n}^\top\mathbf{x}-c\right)^2 \tag{1.1.28.6}$$

$$\implies \|\mathbf{n}\|^2\left(\mathbf{x}^\top\mathbf{x}-2\mathbf{F}^\top\mathbf{x}+\|\mathbf{F}\|^2\right) = e^2\left(c^2+\left(\mathbf{n}^\top\mathbf{x}\right)^2-2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.28.7}$$

$$= e^2\left(c^2+\left(\mathbf{x}^\top\mathbf{n}\mathbf{n}^\top\mathbf{x}\right)-2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.28.8}$$

which can be expressed as (1.1.28.1) after simplification.

1.1.29 The polar form of a conic section is given as:

$$r = \frac{ep}{1+e\cos(\theta-\theta_0)} \tag{1.1.29.1}$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.28.1), the general equation of a conic in matrix form is given by

$$\mathbf{x}^\top\mathbf{V}\mathbf{x}+2\mathbf{u}^\top\mathbf{x}+f=0 \tag{1.1.29.2}$$

For a conic with a focus at the origin ($\mathbf{F}=\mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix}\cos\theta_0\\ \sin\theta_0\end{pmatrix}$, eccentricity e, and directrix distance parameter $c=-p$, substituting into (1.1.28.2)–(1.1.28.4) yields:

$$\mathbf{V} = \mathbf{I}-e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.29.3}$$

$$\mathbf{u} = -pe^2\mathbf{n}, \tag{1.1.29.4}$$

$$f = -p^2e^2 \tag{1.1.29.5}$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r\mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} \tag{1.1.29.6}$$

Substituting into (1.1.29.2) yields

$$r^2\left(\mathbf{r}(\theta)^\top\mathbf{V}\mathbf{r}(\theta)\right)+2r\left(\mathbf{u}^\top\mathbf{r}(\theta)\right)+f=0 \tag{1.1.29.7}$$

Evaluating the inner products:

$$\mathbf{n}^{\top}\mathbf{r}\left(\theta\right)=\begin{pmatrix}\cos\theta_0 & \sin\theta_0\end{pmatrix}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}=\cos\left(\theta-\theta_0\right) \tag{1.1.29.8}$$

$$\mathbf{r}\left(\theta\right)^{\top}\mathbf{V}\mathbf{r}\left(\theta\right)=\mathbf{r}\left(\theta\right)^{\top}\left(\mathbf{I}-e^2\mathbf{n}\mathbf{n}^{\top}\right)\mathbf{r}\left(\theta\right)=1-e^2\cos^2\left(\theta-\theta_0\right) \tag{1.1.29.9}$$

$$\mathbf{u}^{\top}\mathbf{r}\left(\theta\right)=-pe^2\cos\left(\theta-\theta_0\right) \tag{1.1.29.10}$$

Substituting these into (1.1.29.7):

$$r^2\left[1-e^2\cos^2\left(\theta-\theta_0\right)\right]-2rpe^2\cos\left(\theta-\theta_0\right)-p^2e^2=0 \tag{1.1.29.11}$$

Factoring the quadratic expression:

$$\left[r\left(1+e\cos\left(\theta-\theta_0\right)\right)-ep\right]\left[r\left(1-e\cos\left(\theta-\theta_0\right)\right)+ep\right]=0 \tag{1.1.29.12}$$

Taking the positive real root for radial distance $r>0$:

$$r\left(1+e\cos\left(\theta-\theta_0\right)\right)=ep \tag{1.1.29.13}$$

yielding (1.1.29.1)

1.1.30 Gravitational parameter of a body μ is defined as

$$\mu=GM \tag{1.1.30.1}$$

where M is the mass of the body

1.1.31 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h=\left\|\mathbf{r}\times\mathbf{v}\right\| \tag{1.1.31.1}$$

1.1.32 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon=\frac{v^2}{2}-\frac{\mu}{r} \tag{1.1.32.1}$$

where

$$r=\left\|\mathbf{r}\right\|,v=\left\|\mathbf{v}\right\| \tag{1.1.32.2}$$

1.1.33 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a}=-\frac{\mu}{r^3}\mathbf{r} \tag{1.1.33.1}$$

1.1.34 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r}\times\mathbf{a}=\mathbf{r}\times\left(-\frac{\mu}{r^3}\mathbf{r}\right)=\mathbf{0} \tag{1.1.34.1}$$

From (1.1.34.1)

$$\frac{d}{dt}(\mathbf{r}\times\mathbf{v}) = \mathbf{v}\times\mathbf{v} + \mathbf{r}\times\mathbf{a} = \mathbf{0} \tag{1.1.34.2}$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r}\times\mathbf{v} = \text{constant} \tag{1.1.34.3}$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a}\times\mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right)\times(\mathbf{r}\times\mathbf{v}) \tag{1.1.34.4}$$

$$= -\frac{\mu}{r^3}\left[(\mathbf{r}\cdot\mathbf{v})\,\mathbf{r} - (\mathbf{r}\cdot\mathbf{r})\,\mathbf{v}\right] \tag{1.1.34.5}$$

$$= \frac{\mu}{r^3}\left[r^2\mathbf{v} - (\mathbf{r}\cdot\mathbf{v})\,\mathbf{r}\right] \tag{1.1.34.6}$$

Using $\mathbf{r}\cdot\mathbf{v} = r\frac{dr}{dt}$, this simplifies to:

$$\mathbf{a}\times\mathbf{h} = \mu\left[\frac{\mathbf{v}}{r} - \frac{1}{r^2}\frac{dr}{dt}\mathbf{r}\right] = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right) \tag{1.1.34.7}$$

Integrating with respect to time t:

$$\mathbf{v}\times\mathbf{h} = \mu\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.34.8}$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.34.8) with $\mathbf{r}$:

$$\mathbf{r}\cdot(\mathbf{v}\times\mathbf{h}) = \mu\mathbf{r}\cdot\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.34.9}$$

$$\implies (\mathbf{r}\times\mathbf{v})\cdot\mathbf{h} = \mu\left(r + rC\cos\theta\right) \tag{1.1.34.10}$$

$$\implies \|\mathbf{h}\|^2 = \mu r\left(1 + C\cos\theta\right) \tag{1.1.34.11}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C\cos\theta} = \frac{ep}{1 + e\cos\theta} \tag{1.1.34.12}$$

This is the same equation as (1.1.29.1) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \tag{1.1.34.13}$$

$$\mathbf{e} = \mathbf{C} \tag{1.1.34.14}$$

1.1.35 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.35.1}$$

Solution: From (1.1.34.8) and (1.1.34.14),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \tag{1.1.35.2}$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \tag{1.1.35.3}$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r}\left[\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})\right] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \tag{1.1.35.4}$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \tag{1.1.35.5}$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \tag{1.1.35.6}$$

Substituting these back into (1.1.35.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \tag{1.1.35.7}$$

$$= 1 + \frac{2h^2}{\mu^2}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) \tag{1.1.35.8}$$

Substituting (1.1.32.1) into (1.1.35.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.35.9}$$

1.1.36 Consider the system A system comprising a bar, spring and mass is shown in the figure Fig. 1.1.36.1 The bar, having negligible mass, is made of a material having Young's modulus E, cross-sectional area A and length L. The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

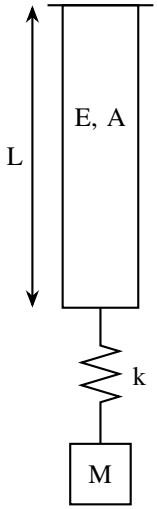


Fig. 1.1.36.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \tag{1.1.36.1}$$

1.1.37 Find the overall spring constant in Fig. 1.1.36.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \tag{1.1.37.1}$$

$$\implies k_{eq} = \frac{k_b k}{k_b + k} \tag{1.1.37.2}$$

where k_t is the overall spring constant.

1.1.38 Show that the equation for motion for the mass in Fig. 1.1.38.1 is

$$\frac{d^2x}{dt^2} = -\frac{k}{M}x \tag{1.1.38.1}$$

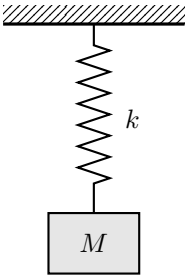


Fig. 1.1.38.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st}.Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \tag{1.1.38.2}$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M\frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \tag{1.1.38.3}$$

$$\tag{1.1.38.4}$$

yielding (1.1.38.1), upon substituting from (1.1.38.2).

2. BT 2025

A. *Formulae*

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.1.1.1}$$

2.1.2 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.2.1}$$

2.1.3 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa\mathbf{m} \tag{2.1.3.1}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0 \tag{2.1.3.2}$$

2.1.4 A circle with center $\mathbf{c}$ and radius r, can be expressed as (1.1.28.1) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{2.1.4.1}$$

Solution: The equation of a circle is

$$\|\mathbf{x} - \mathbf{c}\|^2 = r^2 \tag{2.1.4.2}$$

$$(\mathbf{x} - \mathbf{c})^\top (\mathbf{x} - \mathbf{c}) = r^2 \tag{2.1.4.3}$$

$$\mathbf{x}^\top \mathbf{x} - 2\mathbf{c}^\top \mathbf{x} + \mathbf{c}^\top \mathbf{c} - r^2 = 0 \tag{2.1.4.4}$$

yielding (2.1.4.1) upon comparing with (1.1.28.1)

2.1.5 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{X} = \begin{pmatrix} \mathbf{h} & \mathbf{m} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \tag{2.1.5.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top (\mathbf{V}\mathbf{h}+\mathbf{u})\right]^2 - g(\mathbf{h})\left(\mathbf{m}^\top \mathbf{V}\mathbf{m}\right)}}{\mathbf{m}^\top \mathbf{V}\mathbf{m}}, i = 1, 2. \tag{2.1.5.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

2.1.6 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu\mathbf{V}_2)\mathbf{x} + 2(\mathbf{u}_1 + \mu\mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.1.6.1}$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\begin{vmatrix} \mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2 \\ (\mathbf{u}_1 + \mu\mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{2.1.6.2}$$

2.1.7 For a differentiable function $f(x)$,

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \tag{2.1.7.1}$$

2.1.8 (Gradient Descent Optimization) For the convex optimization problem

$$\min_x f(x), \tag{2.1.8.1}$$

$$x_{k+1} = x_k - \eta f'(x_k) \tag{2.1.8.2}$$

is the iterative parameter update with learning rate $\eta > 0$.

2.1.9 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{2.1.9.1}$$

2.1.10 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.10.1}$$

is consistent iff

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{2.1.10.2}$$

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1, 2, 3, 4, 5, 6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF):

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.11.1}$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete

convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$:

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n-k \mid X_2 = k)\, p_{X_2}(k) \tag{2.1.11.2}$$

$$= \sum_k p_{X_1}(n-k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{2.1.11.3}$$

- Substituting (2.1.11.1) into (2.1.11.3) yields the triangular PMF

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \le n \le 7 \\ \frac{13-n}{36}, & 7 < n \le 12 \\ 0, & n > 12 \end{cases} \tag{2.1.11.4}$$

The following code verifies (2.1.11.4) in Fig. 2.1.11.1.

codes/btq35/dice.py

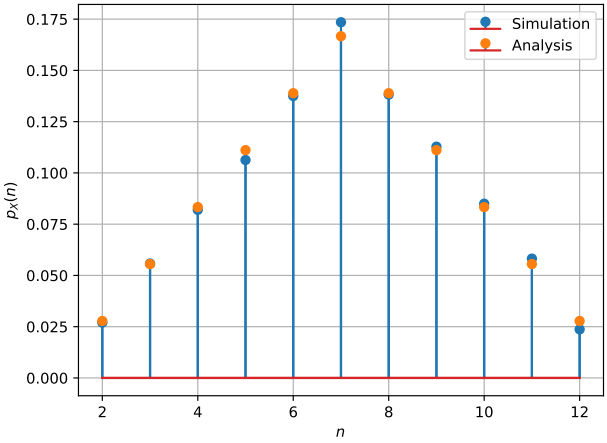


Fig. 2.1.11.1. PMF of the sum of two fair dice: Simulation vs Analysis

2.1.12 (Unit Step Response of an RC Circuit) Consider a series RC circuit excited by a step input $v_i(t) = V_0 u(t)$ where V_0 is a constant. Applying KVL around the loop,

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.1}$$

yielding

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.2}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.3}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.4}$$

$$\implies v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.5}$$

2.1.13 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ iff

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.13.1}$$

2.1.14 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h\to 0^-} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h\to 0^+} \frac{f(x_0+h) - f(x_0)}{h} = f'\left(x_0^+\right) \tag{2.1.14.1}$$

2.1.15 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$

$$\lim_{x\to x_0}\left[f(x) - f(x_0)\right] = \lim_{x\to x_0}\left[\frac{f(x) - f(x_0)}{x - x_0}(x - x_0)\right] = f'(x_0)\cdot 0 = 0 \tag{2.1.15.1}$$

2.1.16 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \le a \\ f_2(x), & x > a \end{cases} \tag{2.1.16.1}$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x\to a^-} f_1(x) = \lim_{x\to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.16.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f_1'(a) = f_2'(a) \tag{2.1.16.3}$$

2.1.17 Representation of Linear Balance Systems as Matrix Equations

- A general system of m linear equations in n variables $x_1, x_2, \ldots, x_n$:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \quad i = 1, 2, \ldots, m \tag{2.1.17.1}$$

can be expressed in matrix-vector form:

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.17.2}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^n$ is the state vector, and $\mathbf{b} \in \mathbb{R}^m$ is the constant vector:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \tag{2.1.17.3}$$

- In multi-component conservation and stoichiometric balances, the conservation of elemental species across reactant and product streams is represented by:

$$\mathbf{A}_{\text{reactants}}\mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{products}}\mathbf{x}_{\text{products}} \tag{2.1.17.4}$$

Combining both sides into a single stoichiometric matrix yields the homogeneous system:

$$\left[\mathbf{A}_{\text{reactants}} \quad -\mathbf{A}_{\text{products}}\right]\mathbf{x} = \mathbf{0} \tag{2.1.17.5}$$

When boundary parameters or specific yields fix certain degrees of freedom (such as basis mole values and yield coefficients), the homogeneous system reduces to a deterministic non-homogeneous matrix equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ that can be solved directly via row operations or symbolic matrix inversion.

2.1.18 Solution of First-Order Recurrence Relations via Z-Transform

- The bilateral Z-transform of a discrete-time sequence $x[n]$ is defined as:

$$X(z) = \mathcal{Z}[x[n]] = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \tag{2.1.18.1}$$

over the Region of Convergence (ROC) where the series converges absolutely.

- Discrete Unit Step Sequence: The discrete unit step function $u[n]$ is defined as:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.1.18.2}$$

Its forward Z-transform is given by:

$$\mathcal{Z}[u[n]] = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \tag{2.1.18.3}$$

- Time-Shift Property: Shifting a sequence by an integer delay k corresponds to multiplication by z^{-k} in the Z-domain:

$$\mathcal{Z}[x[n-k]] = z^{-k} X(z) \tag{2.1.18.4}$$

Consequently, delaying the discrete unit step by one sample yields:

$$\mathcal{Z}\left[u\left[n-1\right]\right]=\frac{z^{-1}}{1-z^{-1}},\quad |z|>1 \tag{2.1.18.5}$$

- Standard Inverse Transform Pair: For any scalar constant α:

$$\mathcal{Z}^{-1}\left[\frac{1}{1-\alpha z^{-1}}\right]=\alpha^{n}u\left[n\right],\quad |z|>|\alpha| \tag{2.1.18.6}$$

2.1.19 Relation Between Eigenvalues and Trace of a Matrix

- Let $\mathbf{A}\in\mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1,\lambda_2,\ldots,\lambda_n$. The characteristic polynomial of $\mathbf{A}$ is defined by:

$$P\left(\lambda\right)=\left|\lambda\mathbf{I}-\mathbf{A}\right| \tag{2.1.19.1}$$

- Factoring the polynomial in terms of its n characteristic roots:

$$P\left(\lambda\right)=\prod_{i=1}^{n}\left(\lambda-\lambda_i\right)=\lambda^{n}-\left(\sum_{i=1}^{n}\lambda_i\right)\lambda^{n-1}+\cdots+\left(-1\right)^{n}\prod_{i=1}^{n}\lambda_i \tag{2.1.19.2}$$

- Evaluating the determinant $\left|\lambda\mathbf{I}-\mathbf{A}\right|$ using cofactor expansion, the only term that contributes to λ^{n} and λ^{n-1} is the product of the principal diagonal entries:

$$\prod_{i=1}^{n}\left(\lambda-a_{ii}\right)=\lambda^{n}-\left(\sum_{i=1}^{n}a_{ii}\right)\lambda^{n-1}+\ldots \tag{2.1.19.3}$$

  where a_{ii} denotes the diagonal elements of $\mathbf{A}$.
- Equating the coefficient of λ^{n-1} from (2.1.19.2) and (2.1.19.3):

$$\sum_{i=1}^{n}\lambda_i=\sum_{i=1}^{n}a_{ii}=\operatorname{tr}\left(\mathbf{A}\right) \tag{2.1.19.4}$$

  Hence, the sum of all eigenvalues of a square matrix is always equal to its trace.

B. Problems

2.2.1 The sum of the following infinite series is:

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\cdots \tag{2.2.1.1}$$

(BT 2025)

a) π b) $1+e$ c) $e-1$ d) e

Solution: (2.2.1.1) can be expressed as

$$\sum_{n=1}^{\infty}\frac{1}{n!}=\sum_{n=0}^{\infty}\frac{1}{n!}-1=e-1 \tag{2.2.1.2}$$

upon substituting from (2.1.1.1) The following C code verifies the series sum numerically.

codes/btq3/btq3.c

2.2.2 A circle with center at $(x,y)=(0.5,0)$ and radius $r_1=0.5$ intersects with another circle with center at $(x,y)=(1,1)$ and radius $r_2=1$ at two points. One of the points of intersection (x,y) is: (BT 2025)

a) $(0,0)$ b) $(0.2,0.4)$ c) $(0.5,0.5)$ d) $(1,2)$

Solution: From (2.1.4.1), the circle parameters are

$$,\mathbf{c}_1=\begin{pmatrix}0.5\\0\end{pmatrix},r_1=0.5,\quad \mathbf{u}_1=-\mathbf{c}_1=\begin{pmatrix}-0.5\\0\end{pmatrix},\quad f_1=\|\mathbf{c}_1\|^2-r_1^2=0 \tag{2.2.2.1}$$

$$\mathbf{c}_2=\begin{pmatrix}1\\1\end{pmatrix},\ r_2=1,\quad \mathbf{u}_2=-\mathbf{c}_2=\begin{pmatrix}-1\\-1\end{pmatrix},\quad f_2=\|\mathbf{c}_2\|^2-r_2^2=1 \tag{2.2.2.2}$$

Substituting the above in (2.1.6.2),

$$\begin{vmatrix}1+\mu & 0 & -0.5-\mu\\ 0 & 1+\mu & -\mu\\ -0.5-\mu & -\mu & \mu\end{vmatrix}=0 \tag{2.2.2.3}$$

$$\implies (1+\mu)\left(-\mu^2-\frac{1}{4}\right)=0 \tag{2.2.2.4}$$

$$\text{or, } \mu=-1 \tag{2.2.2.5}$$

ignoring imaginary roots. Substituting $\mu=-1$ in (2.1.6.1),

$$2\left(\mathbf{u}_1-\mathbf{u}_2\right)^{\top}\mathbf{x}+(f_1-f_2)=0 \tag{2.2.2.6}$$

$$\implies \begin{pmatrix}1 & 2\end{pmatrix}\mathbf{x}=1 \tag{2.2.2.7}$$

From (2.1.2.1) and (2.1.3.1),

$$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.2.2.8}$$

Substituting the above along with the first circle paramters from (2.2.2.1) in (2.1.5.2),

$$\mathbf{m}^\top \mathbf{V}_1 \mathbf{m} = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 5 \tag{2.2.2.9}$$

$$\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \tag{2.2.2.10}$$

$$\mathbf{m}^\top \left(\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 \right) = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} = 1 \tag{2.2.2.11}$$

$$g\left(\mathbf{h}\right) = \mathbf{h}^\top \mathbf{V}_1 \mathbf{h} + 2\mathbf{u}_1^\top \mathbf{h} + f_1 = 1 - 1 + 0 = 0 \tag{2.2.2.12}$$

yielding

$$\kappa = \frac{-1 \pm \sqrt{1^2 - 0}}{5} \implies \kappa_1 = 0, \kappa_2 = -\frac{2}{5} \tag{2.2.2.13}$$

$\therefore$ the points of intersection are

$$\mathbf{x}_1 = \mathbf{h} + 0\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.2.2.14}$$

$$\mathbf{x}_2 = \mathbf{h} - \frac{2}{5}\mathbf{m} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix} \tag{2.2.2.15}$$

Hence, one of the points of intersection is $(0.2, 0.4)$ The following code verifies this in Fig. 2.2.2.1.

codes/btq9/circleProblem1.py

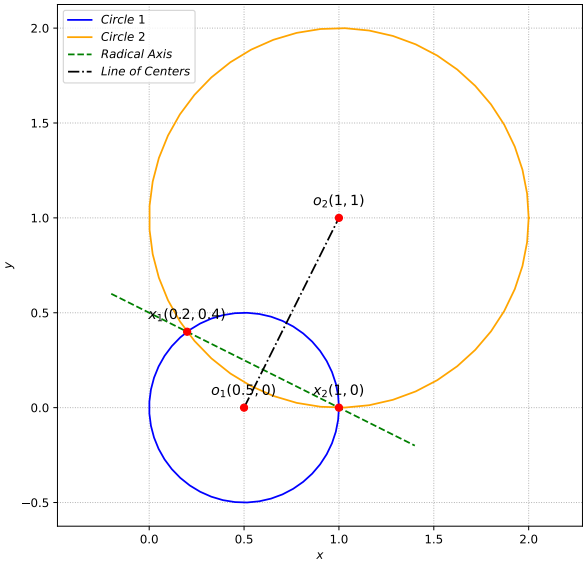


Fig. 2.2.2.1. Intersection of the two circles and the radical axis

2.2.3 Let $y\,(t)$ be a bacterial population whose growth is given by

$$\frac{dy}{dt} = \lambda\,(y + 2) \tag{2.2.3.1}$$

where λ is the growth rate constant. If $y\,(0) = 1$ and $y\,(1) = 4$, then the value of λ is (BT 2025)

a) $\ln 2$ b) $\ln 3$ c) $\ln 4$ d) $\ln 6$

Solution: Rearranging the differential equation into standard linear form

$$\frac{dy}{dt} - \lambda y = 2\lambda u(t) \tag{2.2.3.2}$$

Taking the unilateral Laplace transform on both sides using (1.1.6.1) and (1.1.10.1):

$$sY(s) - y\,(0) - \lambda Y(s) = \frac{2\lambda}{s} \tag{2.2.3.3}$$

Substituting the initial condition $y(0) = 1$,

$$(s-\lambda)\,Y(s) - 1 = \frac{2\lambda}{s} \tag{2.2.3.4}$$

$$\implies Y(s) = \frac{s+2\lambda}{s\,(s-\lambda)} \tag{2.2.3.5}$$

$$= -\frac{2}{s} + \frac{3}{s-\lambda} \tag{2.2.3.6}$$

Applying the inverse Laplace transform using (1.1.10.1) and (1.1.8.1),

$$y(t) = 3e^{\lambda t} - 2 \tag{2.2.3.7}$$

$$\implies y(1) = 4 = 3e^{\lambda} - 2 \tag{2.2.3.8}$$

$$\text{or, } \lambda = \ln 2 \tag{2.2.3.9}$$

From (1.1.13.3), with step size $h = 0.01$ and $\lambda = \ln 2$, the following difference equation is obtained.

$$y_{n+1} = y_n + 0.01\left[\ln(2)\,(y_n + 2)\right], \quad y_0 = 1 \tag{2.2.3.10}$$

The following codes are used to generate Fig. 2.2.3.1.

codes/btq13/bac.py

codes/btq13/bac.c

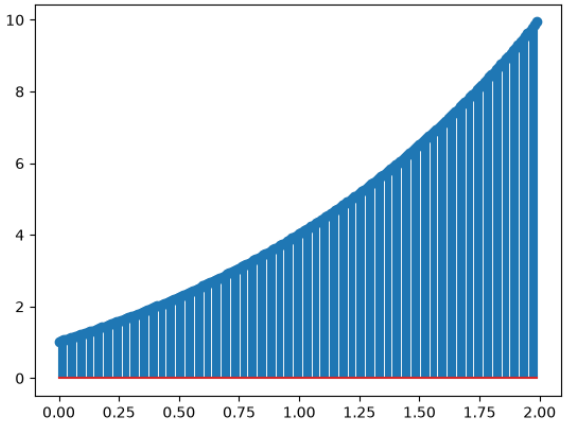


Fig. 2.2.3.1. Euler's Forward Difference Simulation of Bacterial Growth

2.2.4 The minimum value of the function

$$f\left(x\right)=x+\frac{4}{x} \tag{2.2.4.1}$$

for $x>0$ is: (BT 2025)

a) 1 b) 2 c) 3 d) 4

Solution: To locate the critical points of $f\left(x\right)$, we set the first derivative to zero:

$$f'\left(x\right)=\frac{d}{dx}\left(x+\frac{4}{x}\right)=1-\frac{4}{x^2}=0 \tag{2.2.4.2}$$

$$1=\frac{4}{x^2}\implies x^2=4 \tag{2.2.4.3}$$

Since $x>0$, the only admissible critical point is:

$$x=2 \tag{2.2.4.4}$$

From (2.1.7.1)

$$f''\left(x\right)=\frac{d^2}{dx^2}\left(x+\frac{4}{x}\right)=\frac{8}{x^3}>0,\quad x\in\left(0,\infty\right) \tag{2.2.4.5}$$

Thus, $f\left(x\right)$ is strictly convex in the above interval. This is verified by the following code in Fig. 2.2.4.1.

codes/btq14/min.py

Applying the first-order update relation from (2.1.8.2) with learning rate $\eta=0.3$:

$$x_{k+1}=x_k-0.3\left(1-\frac{4}{x_k^2}\right) \tag{2.2.4.6}$$

Starting from the initial guess $x_0=5$, the sequence converges to $x^*=2$, yielding $f\left(x^*\right)=4$. The following code accomplishes this.

codes/btq14/min.c

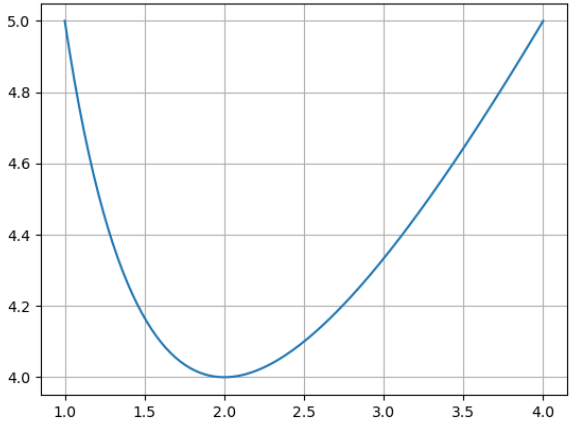


Fig. 2.2.4.1. Plot of $f(x) = x + \frac{4}{x}$ indicating the minimum at $(2, 4)$

2.2.5 Consider a nonlinear algebraic equation, $e^x - 2 = 0$. Using the Newton-Raphson method, with the initial guess of $x_0 = 1$, the approximated value of the root of the equation after one iteration is _____ (round off to two decimal places). (BT 2025)
Solution: Substituting

$$f(x) = e^x - 2 \tag{2.2.5.1}$$
$$f'(x) = e^x \tag{2.2.5.2}$$

into (2.1.9.1),

$$x_{n+1} = x_n - \frac{e^{x_n} - 2}{e^{x_n}} \tag{2.2.5.3}$$
$$= x_n - \left(1 - \frac{2}{e^{x_n}}\right) \tag{2.2.5.4}$$
$$= x_n - 1 + 2e^{-x_n} \tag{2.2.5.5}$$

Given the initial guess $x_0 = 1$, the first iteration $(n = 0)$ gives

$$x_1 \approx x_0 - 1 + 2e^{-x_0} \tag{2.2.5.6}$$
$$= 1 - 1 + 2e^{-1} \tag{2.2.5.7}$$
$$= \frac{2}{e} \tag{2.2.5.8}$$

From the following code,

$$x_4 = 0.693147 \approx \ln 2 \tag{2.2.5.9}$$

which is the exact solution.

codes/btq33/raph.c

The following code generates Fig. 2.2.5.1.

codes/btq33/raph.py

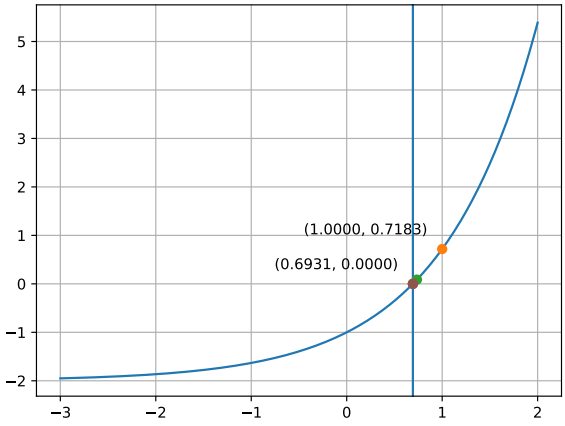


Fig. 2.2.5.1. Convergence of the Newton-Raphson method to $x = \ln 2$

2.2.6 The value of k, for which the linear equations $2x + 3y = 6$ and $4x + 6y = 3k$ have at least one solution, is _____ (answer in integer). (BT 2025)

Solution: Expressing in matrix form $\mathbf{Ax} = \mathbf{b}$,

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 3k \end{pmatrix} \tag{2.2.6.1}$$

The augmented matrix is

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array}\right) \tag{2.2.6.2}$$

Applying Gaussian elimination using elementary row operations,

$$\left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3k-12 \end{array}\right) \tag{2.2.6.3}$$

$\because \text{rank}\,(\mathbf{A}) = 1$, from the above, (2.2.6.1) is consistent if

$$\text{rank}\,\big((\mathbf{A} \mid \mathbf{b})\big) = \text{rank}\,(\mathbf{A}) = 1 \tag{2.2.6.4}$$

$$\implies 3k - 12 = 0 \tag{2.2.6.5}$$

$$\text{or, } k = 4 \tag{2.2.6.6}$$

The following code verifies the result by generating Fig. 2.2.6.1.

codes/btq34/bt34.py

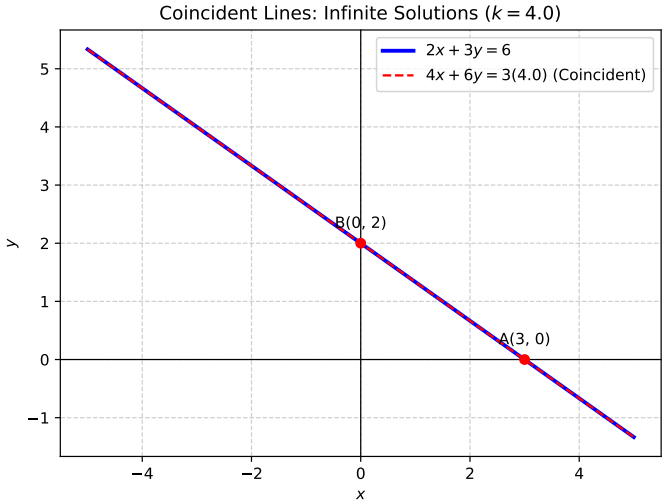


Fig. 2.2.6.1. Coincident lines representing infinitely many solutions at $k = 4$

2.2.7 Two fair six-sided dice, with sides numbered 1 to 6, are thrown once. The probability of getting 7 as the sum of the numbers on the top side of the dice is _____. (BT 2025)

Solution: Let X_1 and X_2 represent the scores on the two dice, and let $X = X_1 + X_2$ denote their sum. From the triangular distribution derived in (2.1.11.4),

$$p_X(n) = \frac{n-1}{36}, \quad 2 \le n \le 7 \tag{2.2.7.1}$$

$$\implies \Pr(X = 7) = \frac{1}{6} \tag{2.2.7.2}$$

2.2.8 A thermometer measuring body temperature follows a first-order response with a time constant of 40 seconds. The instrument will reach 95% of its steady-state output at _____ seconds (round off to the nearest integer). (BT 2025)

a) 60 b) 80 c) 120 d) 160

Solution: From (2.1.12.5),

$$\frac{v_o(t)}{V_0} = \left(1 - e^{-\frac{t}{RC}}\right) u(t) \tag{2.2.8.1}$$

$$\implies 0.95 = \left(1 - e^{-\frac{t}{40}}\right) \tag{2.2.8.2}$$

$$\implies t = 40 \ln(20) \approx 120 \text{ s} \tag{2.2.8.3}$$

The following code generates Fig. 2.2.8.1. for verification.

codes/btq42/plot.py

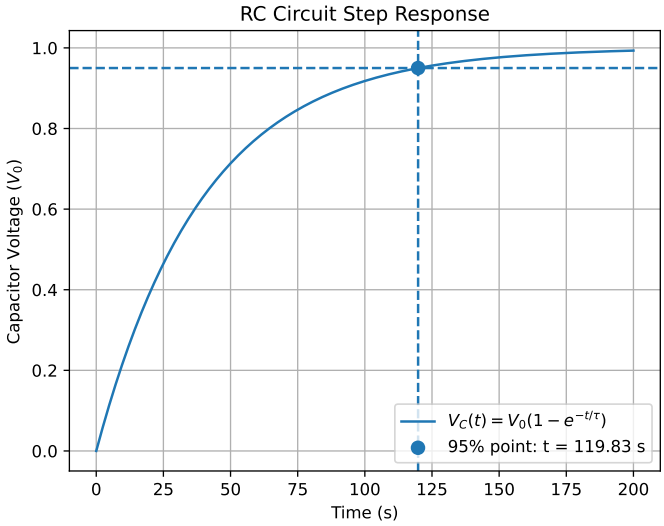


Fig. 2.2.8.1. Step response of the first-order system with $\tau = 40$ s

2.2.9 The output $y(t)$ of a first-order process is governed by the following differential equation

$$\tau_p \frac{dy}{dt} + y = K_p f(t) \tag{2.2.9.1}$$

where τ_p is a non-zero time constant, K_p is the gain and $f(t)$ is the input with $f(0) = 0$. Assume $y(0) = 0$. The transfer function for this process is (consider s as the independent variable in the Laplace domain) (BT 2025)

a) $\frac{K_p}{\tau_p s+1}$ b) $\frac{\tau_p}{K_p s+1}$ c) $\frac{\tau_p}{K_p(s+1)}$ d) $\frac{K_p}{\tau_p(s+1)}$

Solution: In the s-domain, (2.2.9.1) becomes

$$\tau_p s Y(s) + Y(s) = K_p F(s) \tag{2.2.9.2}$$

$$\implies \frac{Y(s)}{F(s)} = \frac{K_p}{\tau_p s + 1} \tag{2.2.9.3}$$

2.2.10 Let m and n be fixed real numbers. If the function $y(t) = C_1 e^t + C_2 e^{-t}$ is a solution of

$$\frac{d^2y}{dt^2} + m\frac{dy}{dt} + ny = 0 \tag{2.2.10.1}$$

for any constants C_1 and C_2, then $m + n$ is equal to (BT 2025)

a) -2 b) -1 c) 0 d) 1

Solution: Substituting

$$\frac{dy}{dt} = C_1 e^{t} - C_2 e^{-t} \tag{2.2.10.2}$$

$$\frac{d^2y}{dt^2} = C_1 e^{t} + C_2 e^{-t} \tag{2.2.10.3}$$

Substituting y, $\frac{dy}{dt}$, and $\frac{d^2y}{dt^2}$ into (2.2.10.1),

$$\left(C_1 e^{t} + C_2 e^{-t}\right) + m\left(C_1 e^{t} - C_2 e^{-t}\right) + n\left(C_1 e^{t} + C_2 e^{-t}\right) = 0 \tag{2.2.10.4}$$

$$\implies C_1 e^{t}\left(1 + m + n\right) + C_2 e^{-t}\left(1 - m + n\right) = 0 \tag{2.2.10.5}$$

$$\text{or, } m + n = -1 \tag{2.2.10.6}$$

2.2.11 Consider the piecewise function

$$f\left(x\right) = \begin{cases} \sin\left(2x\right), & x > 0 \\ a + bx, & x \leq 0 \end{cases} \tag{2.2.11.1}$$

where a and b are real constants. If $f\left(x\right)$ is differentiable at $x = 0$, then the value of $a + b$ is (BT 2025)

a) 0 b) 1 c) 2 d) 3

Solution: Evaluating the one-sided limits

$$\lim_{x \to 0^+} f\left(x\right) = \lim_{x \to 0^+} \sin\left(2x\right) = \sin\left(0\right) = 0 \tag{2.2.11.2}$$

$$\lim_{x \to 0^-} f\left(x\right) = \lim_{x \to 0^-} \left(a + bx\right) = a + b\left(0\right) = a \tag{2.2.11.3}$$

$$\implies a = 0 \tag{2.2.11.4}$$

from (2.1.15.1) and (2.1.13.1). Also,

$$f'\left(0^+\right) = 2\cos\left(0\right) = 2 \tag{2.2.11.5}$$

$$f'\left(0^-\right) = b \implies b = 2 \tag{2.2.11.6}$$

from (2.1.14.1).

$$\therefore a + b = 2 \tag{2.2.11.7}$$

The following code generates Fig. 2.2.11.1 for verification.

codes/btq46/bt46.py

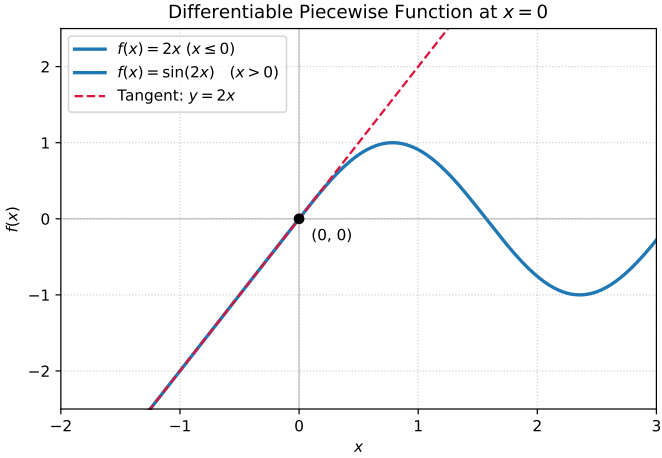


Fig. 2.2.11.1. Differentiable piecewise function and its tangent line $y = 2x$ at $x = 0$

2.2.12 In a fermentation process, each mole of glucose is converted to biomass ($CH_{1.8}O_{0.5}N_{0.2}$), with a biomass yield coefficient of 0.4 C-mol/C-mol, according to the unbalanced equation given below.

$$C_6H_{12}O_6 + NH_3 + O_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + CO_2 + H_2O \tag{2.2.12.1}$$

The moles of oxygen consumption per mole of glucose consumed during fermentation is _____. (BT 2025)

Solution: Let the balanced stoichiometric reaction be represented as:

$$x_1C_6H_{12}O_6 + x_2NH_3 + x_3O_2 \rightarrow x_4CH_{1.8}O_{0.5}N_{0.2} + x_5CO_2 + x_6H_2O \tag{2.2.12.2}$$

where $x_1, x_2, x_3, x_4, x_5, x_6$ are the respective stoichiometric coefficients.
Formulating the elemental conservation balances for C, N, H, and O using exact rational fractions:

$$\text{Carbon (C):} \quad 6x_1 = x_4 + x_5 \tag{2.2.12.3}$$

$$\text{Nitrogen (N):} \quad x_2 = \frac{1}{5}x_4 \tag{2.2.12.4}$$

$$\text{Hydrogen (H):} \quad 12x_1 + 3x_2 = \frac{9}{5}x_4 + 2x_6 \tag{2.2.12.5}$$

$$\text{Oxygen (O):} \quad 6x_1 + 2x_3 = \frac{1}{2}x_4 + 2x_5 + x_6 \tag{2.2.12.6}$$

Writing the elemental balances in matrix form $\mathbf{A}_{\text{LHS}}\mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{RHS}}\mathbf{x}_{\text{products}}$:

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 12 & 3 & 0 \\ 6 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{5} & 0 & 0 \\ \frac{9}{5} & 0 & 2 \\ \frac{1}{2} & 2 & 1 \end{pmatrix} \begin{pmatrix} x_4 \\ x_5 \\ x_6 \end{pmatrix} \tag{2.2.12.7}$$

Constructing the stoichiometric matrix $\mathbf{A} = [\mathbf{A}_{\text{LHS}} \quad -\mathbf{A}_{\text{RHS}}]$:

$$\begin{pmatrix} 6 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & 0 & 0 \\ 12 & 3 & 0 & -\frac{3}{5} & 0 & -2 \\ 6 & 0 & 2 & -\frac{1}{2} & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{2.2.12.8}$$

The biomass yield coefficient is 0.4 C-mol/C-mol $= \frac{2}{5}$ C-mol/C-mol, which provides the relation:

$$x_4 - \frac{2}{5}(6x_1) = 0 \implies x_4 = \frac{12}{5}x_1 \tag{2.2.12.9}$$

Solving the system symbolically in terms of x_4:

$$x_1 = \frac{5}{12}x_4 \tag{2.2.12.10}$$

$$x_2 = \frac{1}{5}x_4 \tag{2.2.12.11}$$

$$x_3 = \frac{29}{20}x_4 \tag{2.2.12.12}$$

$$x_5 = \frac{3}{2}x_4 \tag{2.2.12.13}$$

$$x_6 = \frac{19}{10}x_4 \tag{2.2.12.14}$$

Evaluating with the sample numerical basis $x_4 = 5.0$:

$$x_1 = \frac{25}{12} \approx 2.0833 \tag{2.2.12.15}$$

$$x_2 = 1.0000 \tag{2.2.12.16}$$

$$x_3 = 7.2500 \tag{2.2.12.17}$$

$$x_4 = 5.0000 \tag{2.2.12.18}$$

$$x_5 = 7.5000 \tag{2.2.12.19}$$

$$x_6 = 9.5000 \tag{2.2.12.20}$$

The moles of oxygen consumed per mole of glucose consumed during fermentation is given by:

$$\frac{x_3}{x_1} = \frac{\frac{29}{20}x_4}{\frac{5}{12}x_4} = \frac{29}{20} \times \frac{12}{5} = \frac{87}{25} = 3.48 \tag{2.2.12.21}$$

or using the numerical evaluation:

$$\frac{x_3}{x_1} = \frac{7.2500}{2.0833} = 3.48 \tag{2.2.12.22}$$

Rounding off to two decimal places, the answer is 3.48.
The implementation in SymPy to find solutions is as follows:

codes/btq58/58.py

2.2.13 Let $a_0 = 0$ and define $a_n = \frac{1}{2}(1 + a_{n-1})$ for all positive integers $n \geq 1$. The least value of n for which $|1 - a_n| < \frac{1}{2^{10}}$ is _____. (BT 2025)

Solution: The given recurrence relation is:

$$a_n = \frac{1}{2}(1 + a_{n-1}), \quad n \geq 1, \quad a_0 = 0 \tag{2.2.13.1}$$

Extending the difference equation across all $n \in \mathbb{Z}$ by causal multiplication with $u[n]$ from (2.1.18.2):

$$a_n u[n] - \frac{1}{2} a_{n-1} u[n-1] = \frac{1}{2} u[n-1] \tag{2.2.13.2}$$

Let $A(z) = \mathcal{Z}[a_n u[n]]$. Applying the time-shift property (2.1.18.4) and the delayed step transform (2.1.18.5):

$$A(z) - \frac{1}{2} z^{-1} A(z) = \frac{1}{2} \mathcal{Z}[u[n-1]] \tag{2.2.13.3}$$

$$A(z)\left(1 - \frac{1}{2} z^{-1}\right) = \frac{\frac{1}{2} z^{-1}}{1 - z^{-1}} \tag{2.2.13.4}$$

Solving algebraically for $A(z)$:

$$A(z) = \frac{\frac{1}{2} z^{-1}}{(1 - z^{-1})\left(1 - \frac{1}{2} z^{-1}\right)}, \quad |z| > 1 \tag{2.2.13.5}$$

Performing partial fraction decomposition with $w = z^{-1}$:

$$\frac{\frac{1}{2} w}{(1 - w)\left(1 - \frac{1}{2} w\right)} = \frac{C_1}{1 - w} + \frac{C_2}{1 - \frac{1}{2} w} \tag{2.2.13.6}$$

where

$$C_1 = \left.\frac{\frac{1}{2} w}{1 - \frac{1}{2} w}\right|_{w=1} = \frac{1/2}{1/2} = 1 \tag{2.2.13.7}$$

$$C_2 = \left.\frac{\frac{1}{2} w}{1 - w}\right|_{w=2} = \frac{1}{1 - 2} = -1 \tag{2.2.13.8}$$

Substituting back into $A(z)$:

$$A(z) = \frac{1}{1 - z^{-1}} - \frac{1}{1 - \frac{1}{2} z^{-1}} \tag{2.2.13.9}$$

Applying the inverse Z-transform using (2.1.18.3) and (2.1.18.6):

$$a_n = \left[1 - \left(\frac{1}{2}\right)^n\right] u[n] = 1 - 2^{-n}, \quad n \geq 0 \tag{2.2.13.10}$$

Applying the stopping threshold condition $|1 - a_n| < \frac{1}{2^{10}}$:

$$\left|1 - \left(1 - 2^{-n}\right)\right| < \frac{1}{2^{10}} \tag{2.2.13.11}$$

$$2^{-n} < 2^{-10} \tag{2.2.13.12}$$

$$2^n > 2^{10} \implies n > 10 \tag{2.2.13.13}$$

Since n must be an integer, the minimal integer satisfying the strict inequality is:

$$n = 11 \tag{2.2.13.14}$$

The implementation in C to simulate the recurrence and verify the threshold is as follows:

codes/btq59/ztransform.c

2.2.14 Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix}$. If the eigenvalues of A are -2, 1, and 2, then the value of k is _____. (BT 2025)

Solution: The given matrix is:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix} \tag{2.2.14.1}$$

The eigenvalues of $\mathbf{A}$ are given as:

$$\lambda_1 = -2, \quad \lambda_2 = 1, \quad \lambda_3 = 2 \tag{2.2.14.2}$$

From (2.1.19.4), the sum of the eigenvalues of a square matrix is equal to its trace:

$$\sum_{i=1}^{3} \lambda_i = \operatorname{tr}(\mathbf{A}) \tag{2.2.14.3}$$

Computing the trace of $\mathbf{A}$:

$$\operatorname{tr}(\mathbf{A}) = 1 + k + (-1) = k \tag{2.2.14.4}$$

Computing the sum of the eigenvalues:

$$\sum_{i=1}^{3} \lambda_i = (-2) + 1 + 2 = 1 \tag{2.2.14.5}$$

Equating the two expressions:

$$k = 1 \tag{2.2.14.6}$$

The solution can be verified from the following Python script:

codes/btq63/eigen.py